

CPE-469 Distributed Systems

Programming Assignment 6

Single-Node Banking Server

In this programming assignment, we want to build a single-node banking server, with multiple clients sending various requests.

The bank server has information about the current bank tellers and bank customers.

Bank tellers are special types of clients, that can:

1. Apply bonus (positive percentage)/interest (negative percentage) to an account.
2. Open/close accounts.
3. Freeze/Unfreeze accounts.
4. Charge service fees (negative input value).

Each customer account has: an account id, username, current balance, and an account status (active or frozen). Note that a customer cannot perform any operation on a closed or frozen account.

All customers can:

1. Check balance
2. Deposit money
3. Withdraw money
4. Transfer money to another customer (using account id)

Create a banking server, that supports all these client requests using RPCs. All RPCs are synchronous and return when the operation succeeds or fails. Clients are responsible for retrying in case of failure. The banking server should be able to handle hundreds of requests concurrently.

Use a database (like sqlite) to store the customers information. Before applying any query to the database, write the operation to a logfile. The logfile keeps track of the sequence of all operations performed. You may use a database table for the logfile.

You should submit go project and a README file.